

Detecting Cannabis–Alcohol Substitution in Aggregate Sales Data: Statistical Power and the Limits of Small-Sample Synthetic Control

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State-level alcohol sales are the most common outcome in studies of whether legal cannabis substitutes for alcohol, and those studies mostly report nulls. This paper asks whether such designs could detect substitution of plausible size. Using per-capita ethanol sales and the staggered opening of recreational cannabis retail, I estimate the pooled effect with six estimators; all six return positive-signed nulls between +0.7 and +2.8 percent, and backdated placebos produce negative pseudo-effects of one to two percent where no state had legalized. A design-based power simulation gives two results. First, the jackknife test conventionally used with the primary estimator has near-zero size and rejects nothing at any effect up to twelve percent, because its standard error runs several times the estimator's true sampling dispersion; nulls from this common small-sample configuration carry little weight. Second, even with calibrated inference the minimum detectable effect is 4.5 to 6.4 percent: two-to-three-percent effects fall below what any of these estimators can detect, and a four-percent effect reaches the frontier of only the most sensitive one, leaving substitution in the low single digits at or below reach .

JEL: C21, C23, I12, I18

Keywords: cannabis legalization, alcohol, substitution, synthetic control, statistical power, staggered adoption

The case for cannabis legalization rests in part on the claim that cannabis substitutes for more harmful substances, alcohol being the most common. The social cost of alcohol in the United States is large and well documented (Sacks et al., 2015; Cook, 2007), so even a modest substitution effect would be a meaningful side benefit of legalization, and its absence would remove a common argument in legalization's favor. Since Colorado and Washington opened the first recreational cannabis retail markets in 2014, a staggered rollout across states has generated a

* Data are from the National Institute on Alcohol Abuse and Alcoholism Surveillance Report #122, state cannabis tax and regulatory portals (Colorado Department of Revenue, Washington State Liquor and Cannabis Board, Oregon Liquor and Cannabis Commission, Nevada Cannabis Compliance Board, California Department of Tax and Fee Administration, Massachusetts Cannabis Control Commission), the Bureau of Economic Analysis, the Bureau of Labor Statistics, the Federation of Tax Administrators, and the U.S. Census Bureau. Replication code and a one-command run script are available in the project repository; every number in this paper is generated programmatically from the results files. Claude (Anthropic) was used for code development, and debugging; all research design decisions, data choices, and final text are my own. All errors are mine.

substantial empirical literature on the cannabis–alcohol relationship.

That literature has not reached a consensus. Anderson, Hansen and Rees (2013) find evidence consistent with substitution in traffic fatality data. Baggio, Chong and Kwon (2020) estimate a roughly 12 percent decline in alcohol sales using retail-scanner data and a border-county design. Veligati et al. (2020) finds no effect in aggregate state tax-receipt data, while Calvert and Erickson (2021) find mixed, beverage- and state-specific results in household scanner-panel data, neither clearly substitution nor complementarity. Reviews of the broader evidence conclude that cannabis and alcohol act as both substitutes and complements depending on the population and the margin studied (Subbaraman, 2016; Risso et al., 2020). The studies that find substitution use granular data (scanner panels, individual fatality records), while those built on state-level aggregates tend to find nulls.

This paper examines whether the aggregate nulls reflect the absence of substitution or the limits of the data. The concern is statistical power: with seven candidate treated states, thirty never-treated control states, and twenty annual observations per state, it is not obvious that any estimator can distinguish a true substitution effect of plausible size from noise. If it cannot, the aggregate nulls carry little information either way, and reporting them as evidence against substitution overstates what the design can support.

I proceed in three steps. First, I estimate the pooled effect of recreational retail opening on log per-capita ethanol sales with six estimators: partially-pooled augmented synthetic control (Ben-Michael, Feller and Rothstein, 2022), synthetic difference-in-differences (Arkhangelsky et al., 2021), generalized synthetic control (Xu, 2017), matrix completion (Athey et al., 2021), the group-time estimator of Callaway and Sant’Anna (2021), and two-way fixed effects. Using several estimators rules out the possibility that any single method’s bias profile drives the conclusion. Second, I backdate treatment to every feasible placebo adoption year and re-estimate, which measures the pseudo-effects each method produces in periods when no state had legalized. Third, I run a design-based power simulation: I inject substitution effects of known size into randomly drawn never-treated states using the real staggered adoption dates, re-estimate, and record how often each procedure rejects the null.

The three exercises give a consistent answer. All six estimators return pooled estimates between +0.7 and +2.8 percent, positive-signed and not significant. The backdated placebos show that every estimator produces a spurious effect of roughly -1 to -2 percent in a no-treatment window, so true effects of that size are inside the design’s noise. The power simulation shows that the primary estimator’s jackknife test, the standard inference for this method in applied work, rejects in zero percent of draws at every injected effect up to 12 percent; with a correctly calibrated randomization test, the primary estimator’s minimum detectable effect at 80 percent power is 6.4 percent of per-capita ethanol, roughly 3.2 standard drinks per adult per month. Extending the simulation to every estima-

tor, the smallest minimum detectable effect in the set is 4.5 percent. The effects this literature entertains run from null to the 12 percent that Baggio, Chong and Kwon (2020) estimate from scanner data; effects in the low single digits, the small end of that range, fall below the detection threshold of every estimator considered.

The paper makes two contributions. The first is a reframing of what the aggregate-data literature has established: the nulls in that literature are statements about detectability rather than evidence that substitution is absent, and they do not contradict the granular-data studies that find substitution. The second is methodological and reaches past cannabis. With a handful of treated units, the jackknife standard errors attached to the partially-pooled synthetic control estimator are several times the estimator’s true sampling dispersion, so the test cannot reject any effect of policy-relevant size; a null result from this procedure is close to guaranteed regardless of the true effect. The configuration that produces it, a few treated states matched to many donors over a couple of decades, is common in applied work, and a design-based power simulation makes the problem visible at low cost.

The paper proceeds as follows. Section I describes the outcome data, the treatment, and the first stage. Section II sets out the sample rules, the six estimators, and the inference. Section III presents the pooled and state-level estimates. Section IV reports the in-time placebo evidence. Section V presents the power analysis. Section VI collects robustness checks. Section VII discusses implications, and Section VIII concludes.

I. Data and Treatment

A. Outcome: Per-Capita Ethanol Sales

The outcome is per-capita ethanol consumption, measured in gallons of pure ethanol per person aged 21 and older, from the National Institute on Alcohol Abuse and Alcoholism Surveillance Report #122 (Slater and Alpert, 2025). The series is constructed from state alcohol tax receipts and shipment data, disaggregates into beer, wine, and spirits, and runs through 2023. I take the natural logarithm so that estimates read as percentage effects, and I restrict the primary analysis window to 2000–2019. The start year gives every treated state at least fourteen pre-treatment years for matching; the end year excludes the COVID-19 period, which shifted both the level and the composition of alcohol sales in ways unrelated to cannabis policy. The full 2000–2023 window appears as a labeled extension in Section VI. One measurement caveat applies throughout: the series measures sales, not consumption, so alcohol purchased in one state and consumed in another is attributed to the state of sale. New Hampshire, whose state-run liquor stores and absence of a sales tax draw heavy cross-border purchasing, is the best-known case, and the spillover robustness checks in Section VI address this class of problem.

Table 1 reports pre-treatment summary statistics. The treated states span

baseline drinking levels from roughly 2.6 gallons per adult (California, Washington) to 3.9 (Nevada), which indicates that no single untreated state can serve as a credible comparison. The donor average is computed over 2000–2013 (the pre-treatment window) so that the comparison row covers the same years used for matching.

TABLE 1—PER-CAPITA ETHANOL BY TREATED STATE, PRE-TREATMENT

State	Period	Mean	SD	Min	Max
Colorado	2000–2013	3.086	0.065	2.975	3.183
Washington	2000–2013	2.574	0.083	2.455	2.704
Oregon	2000–2014	2.833	0.146	2.598	3.012
Alaska	2000–2015	3.201	0.179	2.853	3.514
Nevada	2000–2016	3.902	0.163	3.628	4.137
California	2000–2017	2.600	0.052	2.487	2.684
Massachusetts	2000–2017	2.863	0.045	2.756	2.950
Donor average	2000–2013	2.613	0.631	1.509	5.394

Notes: Gallons of pure ethanol per person aged 21 and older, NIAAA Surveillance Report #122. Each treated state’s row covers its own matching period, 2000 through the year before retail opening. The donor row covers 2000–2013, the matching window, across the 30 states that had not legalized recreational cannabis through 2023.

Figure 1 plots log per-capita ethanol for the early legalizers (Colorado and Washington), the later legalizers, and the never-legalized states. The early legalizers sit below the never-legalized average throughout, and the gap widens over time. The drift rules out a simple before-and-after comparison and motivates constructed counterfactuals; it is also, as Section IV shows, the design’s central vulnerability.

B. Treatment Timing and the First Stage

The treatment is the opening of legal recreational retail rather than the ballot measure, because retail access is the point at which consumer behavior can respond. The seven candidate treated states and their retail years are Colorado (2014), Washington (2014), Oregon (2015), Alaska (2016), Nevada (2017), California (2018), and Massachusetts (2018).

A null result is informative only if the treatment actually changed access, so I document the first stage directly with annual legal cannabis sales collected from each state’s tax or regulatory authority.¹ Figure 2 shows per-adult legal sales. Retail opening produced large and fast access changes: Colorado reached roughly

¹Colorado Department of Revenue marijuana sales reports; Washington State Liquor and Cannabis Board sales files (fiscal years); Oregon Liquor and Cannabis Commission legislative reports (rounded annual totals only, as the agency’s dashboard does not export data); Nevada Department of Taxation and Cannabis Compliance Board releases (fiscal years); California Department of Tax and Fee Administration quarterly taxable sales; Massachusetts Cannabis Control Commission open data. Sources, retrieval dates, and caveats for every figure are documented in the replication repository.

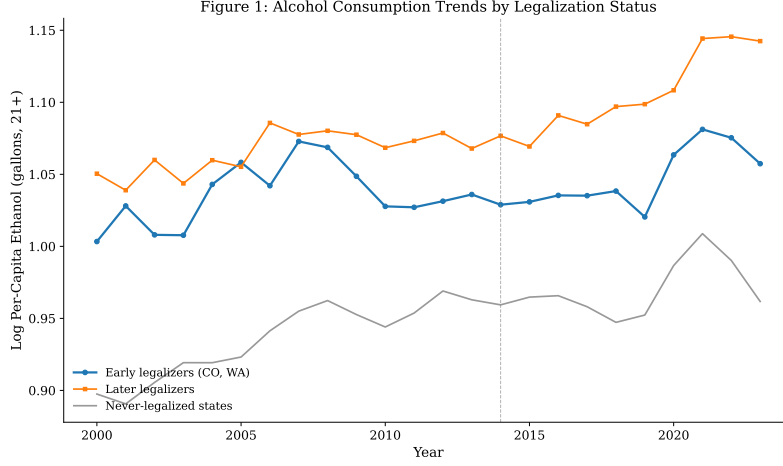


FIGURE 1. ALCOHOL CONSUMPTION TRENDS BY LEGALIZATION STATUS

Notes: Mean log per-capita ethanol by group. Early legalizers are Colorado and Washington; later legalizers are Oregon, Alaska, Nevada, California, and Massachusetts; never-legalized states are the 30 states without recreational retail through 2023. The dashed line marks 2014.

\$213 per adult per year in its first two retail years and Nevada roughly \$236. California is the exception. Its dose at opening was about \$83 per adult, 2.6 times smaller than Colorado’s, because California had operated a near-universal de facto medical access regime since 1996, so the 2018 retail opening changed the legal form of access more than its effective availability. This asymmetry matters for interpreting California’s estimate in Section III.C.

C. Donor Pool and Covariates

The primary donor pool is the 30 states that never legalized recreational cannabis through 2023. I use never-treated states rather than not-yet-treated states because a donor that legalizes later in the sample enters the comparison partially treated, which biases the estimated effect toward zero. A timing-adjusted pool that also admits states legalizing after the 2000–2019 window closes is reported as robustness. The District of Columbia is excluded throughout.

The primary specification matches on the pre-treatment outcome path alone, with no auxiliary covariates. The reason is a result in Kaul et al. (2022): when a synthetic control matches on all pre-treatment outcome lags, covariates receive zero weight, so adding them creates the appearance of covariate adjustment without affecting the estimate. I therefore state the outcome-only specification as the primary one and report a lag-subset specification, odd-year outcome lags plus four covariates measured as the average over the three years before each state’s own treatment, as robustness in Section VI. The covariates are real GDP per capita (Bureau of Economic Analysis), the unemployment rate (Bureau of Labor

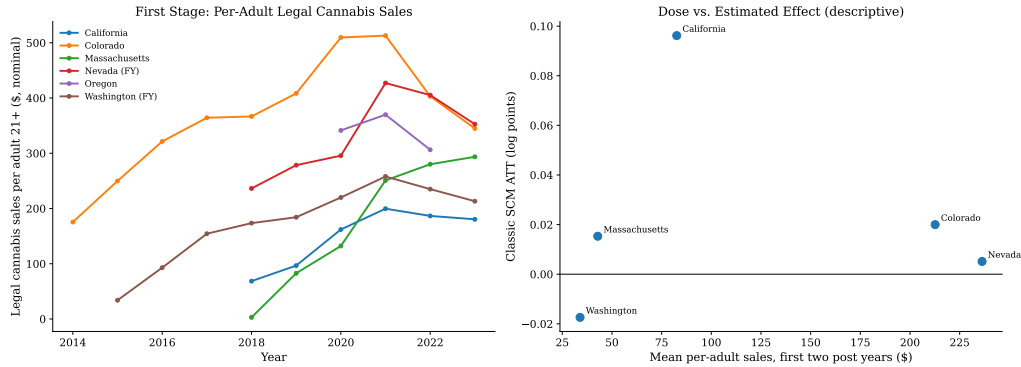


FIGURE 2. FIRST STAGE: LEGAL CANNABIS SALES AND ESTIMATED EFFECTS BY DOSE

Notes: Left: annual legal cannabis sales per adult aged 21 and older, nominal dollars, from state tax and regulatory portals; Washington and Nevada report fiscal years. California’s figure is taxable sales and therefore overstates the change in effective access given pre-existing medical availability. Right: each state’s classic synthetic control estimate against its mean per-adult sales in the first two retail years. The right panel is a five-point diagnostic; no slope is estimated from it.

Statistics), the share of the population aged 20–34 (Census Bureau population estimates), and the beer excise tax (Federation of Tax Administrators), each measured pre-treatment for each state rather than at a fixed baseline year.

II. Empirical Design

A. Pre-Specified Sample Rules

Synthetic control estimates are credible only where the pre-treatment fit is credible (Abadie, 2021), and fit-based sample decisions made after seeing the estimates invite cherry-picking. I therefore fix two inclusion rules before estimating any treatment effect, using only pre-treatment information, and apply them uniformly.

The first rule is a fit-quality screen: a treated state enters the pooled sample only if its classic synthetic control pre-treatment root mean squared prediction error (RMSPE) is no more than twice the median RMSPE of in-space placebos, where each never-treated donor is fitted identically, as if treated at the same date. The placebo benchmark asks whether the method tracks the treated state as well as it tracks states where there is nothing unusual to track; the factor of two allows for sampling variation in the placebo fits while excluding clear failures, and was fixed before estimation. Table 2 reports the screen. Alaska and Nevada fail at 4.3 and 5.1 times the placebo median; the five remaining states pass. The primary sample is therefore California, Colorado, Massachusetts, Oregon, Washington.²

²An alternative normalization, RMSPE relative to the standard deviation of the state’s own pre-period outcome, is shown in the table for transparency. It would also exclude Colorado, not because

The second rule concerns Washington, which privatized liquor sales in June 2012, two years before its cannabis retail opening. A level shift from privatization could be mistaken for a treatment effect. I run a Chow forecast test for a 2012 break in Washington’s donor-demeaned log ethanol over 2000–2013; the test does not reject ($p = 0.17$ for total ethanol), so Washington stays in the sample. Section VI additionally reports Washington on a spirits-excluded outcome, where any privatization artifact would be concentrated.

TABLE 2—PRE-SPECIFIED FIT SCREEN

State	T_0	Pre-RMSPE	Placebo median	Ratio	Own-SD ratio	Included
Colorado	2014	0.0184	0.0126	1.46	0.86	Yes
Washington	2014	0.0149	0.0126	1.19	0.46	Yes
Oregon	2015	0.0150	0.0123	1.22	0.29	Yes
Alaska	2016	0.0521	0.0121	4.29	0.93	No
Nevada	2017	0.0602	0.0118	5.08	1.43	No
California	2018	0.0071	0.0129	0.55	0.35	Yes
Massachusetts	2018	0.0068	0.0129	0.53	0.43	Yes

Notes: Classic synthetic control fitted on the outcome path only, never-treated donors, 2000–2019 window, with treatment at each state’s retail year T_0 . The placebo median is the median pre-RMSPE across the 30 never-treated donors fitted identically at the same T_0 . The inclusion rule, fixed before estimation, is $\text{Ratio} \leq 2$. The own-SD ratio (pre-RMSPE over the standard deviation of the state’s own pre-period log outcome) is reported as a diagnostic only.

B. Six Estimators

All estimators are applied to the same object: the pooled effect of retail opening on log per-capita ethanol over 2000–2019, for the five clean-fit treated states against the never-treated donor pool, with each state’s treatment switching on at its own retail year.

The primary estimator is the partially-pooled staggered augmented synthetic control of Ben-Michael, Feller and Rothstein (2022). It estimates one synthetic control per treated state while shrinking the state-level imbalances toward a pooled target, and it aggregates across adoption cohorts in event time, so it is built for exactly this setting: a few treated units adopting at different dates. Inference is by jackknife over treated units, the package default and the standard in applied work using this method.

I report five alternatives, chosen so that the identifying assumptions differ. Synthetic difference-in-differences (Arkhangelsky et al., 2021) reweights both units and time periods and, unlike synthetic control, allows a level shift between treated and synthetic units. Because the estimator assumes simultaneous adoption, I estimate it separately by adoption cohort on balanced blocks and aggregate by

Colorado fits poorly but because Colorado’s pre-period series is nearly flat, so the denominator is small.

treated post-period cell counts, with placebo-variance inference.³ The generalized synthetic control (Xu, 2017) fits an interactive fixed effects model with the factor number chosen by cross-validation (selected $r = 1$) and parametric-bootstrap inference. Matrix completion (Athey et al., 2021) imputes the counterfactual by nuclear-norm-regularized completion of the untreated outcome matrix. The group-time estimator of Callaway and Sant’Anna (2021), with never-treated controls and multiplier-bootstrap inference, represents the modern difference-in-differences approach and avoids the negative-weighting problems of two-way fixed effects under staggered adoption (Goodman-Bacon, 2021; de Chaisemartin and D’Haultfoe uille, 2020). Plain two-way fixed effects with state-clustered standard errors completes the set as the conventional benchmark.

C. Inference

Each estimator carries its own native inference, reported with replication counts in Table 3. I add three pieces. First, a joint randomization-inference test of the pooled null: I draw 5 pseudo-treated states from the 30 never-treated donors 500 times, assign them the real staggered adoption dates at random, re-estimate the primary estimator, and locate the observed pooled estimate in the resulting null distribution. This test uses the same panel, the same dates, and the same estimator as the actual analysis, so it is the paper’s preferred test of the joint null. Second, for the state-by-state estimates I report prediction intervals from the method of Cattaneo, Feng and Titiunik (2021).⁴ Third, the power simulation of Section V measures what any of these tests can and cannot detect.

III. Results

A. Pooled Estimates

Table 3 and Figure 3 report the pooled estimates. Six estimators spanning a range of identification approaches return pooled estimates between +0.7 and +2.8 percent. Every point estimate is positive, the wrong sign for substitution; every confidence interval covers zero; and the two-sided randomization-inference test of the joint null gives $p = 0.17$ against a null distribution with standard deviation 0.020 log points. Because the point estimates carry the wrong sign for substitution, a one-sided test against substitution would be even less favorable. The null is not specific to any one estimator’s identifying assumptions.

The Callaway–Sant’Anna event study (Figure 4) shows the same picture in event time: post-treatment effects scattered around zero with no trend. The

³The cohort estimates share the donor pool, so treating them as independent in the aggregated standard error is an approximation. The conclusions are insensitive to inflating this standard error substantially.

⁴Permutation inference with 30 donors has a hard p-value floor of $1/31 \approx 0.032$: no treated state can achieve a smaller p-value, so multiple-comparison corrections that require thresholds below the floor cannot be satisfied by construction.

TABLE 3—POOLED EFFECT OF RECREATIONAL RETAIL OPENING: SIX ESTIMATORS

Estimator	ATT	(SE)	Percent	95% CI	Inference
Partially-pooled ASCM (primary)	0.0276	(0.0419)	+2.8	[-0.055, +0.110]	Jackknife
Synthetic difference-in-differences	0.0181	(0.0153)	+1.8	[-0.012, +0.048]	Placebo (200 reps/cohort)
Generalized synthetic control (IFE)	0.0179	(0.0207)	+1.8	[-0.023, +0.058]	Param. bootstrap (500)
Matrix completion	0.0077	(0.0201)	+0.8	[-0.032, +0.047]	Bootstrap (500)
Callaway–Sant’Anna	0.0124	(0.0122)	+1.3	[-0.011, +0.036]	Multiplier bootstrap (999)
Two-way fixed effects	0.0071	(0.0229)	+0.7	[-0.038, +0.052]	Clustered by state

Notes: Outcome is log per-capita ethanol (21+), 2000–2019. Treated states are California, Colorado, Massachusetts, Oregon, Washington (retail years 2014, 2014, 2015, 2018, 2018); donors are the 30 never-treated states. ATT in log points; the percent column is $e^{\text{ATT}} - 1$. Synthetic difference-in-differences is estimated by adoption cohort and aggregated by treated post-period cells. Joint randomization-inference test of the pooled null using the primary estimator: $p = 0.17$ (500 draws).

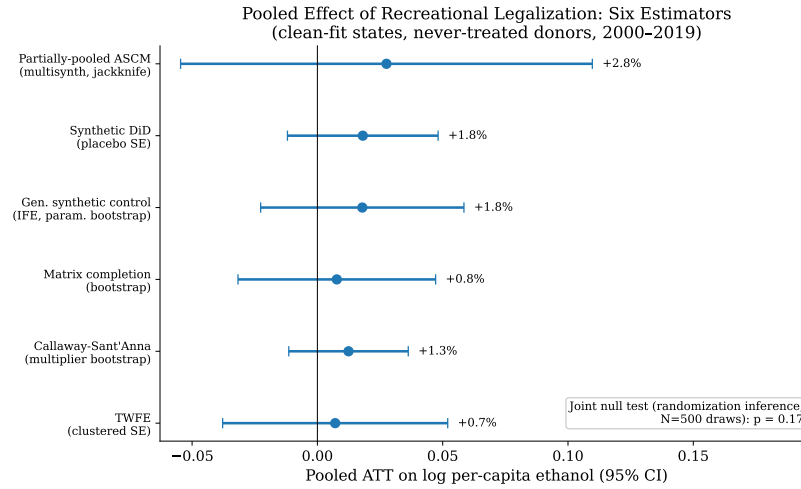


FIGURE 3. POOLED ESTIMATES: SIX ESTIMATORS

Notes: Pooled ATT estimates with 95% confidence intervals from Table 3. The annotation reports the randomization-inference p-value for the joint null under the primary estimator.

pre-treatment coefficients are mostly flat but not perfectly so, which is examined directly in Section IV.

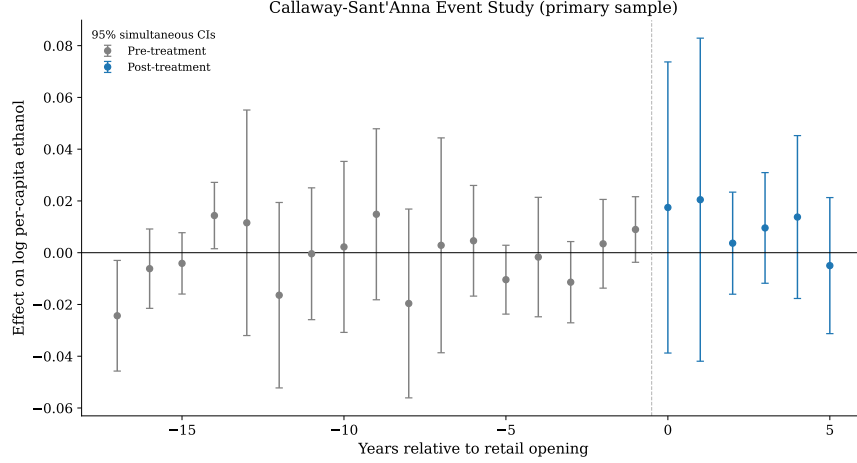


FIGURE 4. CALLAWAY-SANT'ANNA EVENT STUDY

Notes: Group-time average treatment effects aggregated to event time, never-treated control group, with 95% simultaneous confidence intervals from a multiplier bootstrap (999 iterations).

B. State-by-State Estimates

Table 4 reports the state-by-state synthetic control estimates, classic and ridge-augmented, together with prediction intervals for the post-period gaps. Colorado, the longest-exposed and highest-dose state, shows +2.0 percent with an interval which includes zero. Washington is the only state whose point estimate carries the substitution sign (−1.7 percent), and its interval also covers zero. California is the outlier in the other direction and is discussed next.

C. California

California's gap is +10.1 percent, with alcohol sales above its synthetic control, and Figure 5 shows that its prediction interval excludes zero in every post-treatment year; the lower envelope of the interval never falls below +5.2 percent. Massachusetts shows the same pattern at a tenth the magnitude (+1.5 percent). Read causally, this would say that legalization raised drinking in the two largest late-adopting states. There are three reasons why this should not be read causally. First, and independent of the gap's size, California's first-stage dose was the smallest in the sample (Section I.B): the state least treated in substance shows the largest estimated effect, which is the opposite of a dose-response relationship, and the right panel of Figure 2 confirms no negative gradient across states.

TABLE 4—STATE-BY-STATE SYNTHETIC CONTROL ESTIMATES

State	T_0	Pre-RMSPE	Classic	Ridge ASCM	Percent	Prediction interval
Colorado	2014	0.0184	0.0200	0.0200	+2.0	[-0.104, +0.099]
Washington	2014	0.0149	-0.0174	-0.0133	-1.7	[-0.152, +0.117]
Oregon	2015	0.0150	0.0217	0.0217	+2.2	[-0.099, +0.097]
Alaska	2016	0.0521	-0.0137	-0.0135	-1.4	—
Nevada	2017	0.0602	0.0051	0.0319	+0.5	—
California	2018	0.0071	0.0962	0.0962	+10.1	[+0.051, +0.131]
Massachusetts	2018	0.0068	0.0153	0.0153	+1.5	[+0.008, +0.024]

Notes: Classic and ridge-augmented synthetic control, outcome-only match, never-treated donors, 2000–2019. ATT in log points; the percent column applies to the classic estimate. The final column is the envelope of the 90% prediction intervals (Cattaneo, Feng and Titiunik, 2021) for the post-period gap across post years, for clean-fit states. Alaska and Nevada are shown for context only; they fail the fit screen and enter no pooled estimate.

Second, the direction matches what donor contamination produces mechanically: under true substitution, cannabis diffusion into the never-treated donor states (through medical programs, informal markets, and cross-border access) depresses donor alcohol sales, which inflates every treated state’s gap upward, and the bias is largest for the latest adopters because their donors have had the longest exposure.⁵ Third, the in-time placebo distributions of Section IV sit on the negative side of zero, so the regional drift that generates spurious negative gaps cannot explain a large positive one. Contamination plausibly accounts for the positive sign and a few points of the magnitude; the remainder is left unexplained by any substitution story, which is what a noisy single-unit design produces when it occasionally throws off a large wrong-signed gap. I present the result as a caution about what aggregate gaps absorb rather than as a treatment effect.

The interval deserves one further note. California’s prediction interval excludes zero at a lower bound of +5.2 percent, below the pooled minimum detectable effect of 6.4 percent reported in Section V, which looks contradictory but is not, because the two are different objects. The pooled figure is the power to detect a common effect averaged across the five treated states under randomization inference; the interval is a single-unit conformal statement about California’s own gap relative to its own synthetic, and California has the cleanest pre-treatment fit in the sample, which makes its interval tight. Massachusetts makes the same point from the small-effect side: its +1.5 percent gap is a low-single-digit effect whose single-unit interval also excludes zero. Both clean-fit intervals exclude zero in part because a single-unit conformal statement does not absorb the regional drift that the in-time placebos of Section IV show generates one-to-two percent pseudo-effects,

⁵Contamination cannot account for the full gap, however. The synthetic control is a convex combination of donor outcomes, so a uniform downward contamination of donor log-alcohol by d raises the measured gap by about d . Reproducing California’s roughly ten-log-point gap from contamination alone would require donor drinking to have fallen near ten percent relative to its uncontaminated path; since donors are only partially exposed to cannabis, that implies a full-legalization substitution effect well into double digits, larger than any estimate in the literature. Contamination plausibly explains the positive sign and a few points of the gap, not its full size.

so these single-unit intervals are not the detectability benchmark; the pooled randomization power is. Resting on two post-treatment years (2018 and 2019), California’s interval is very short.

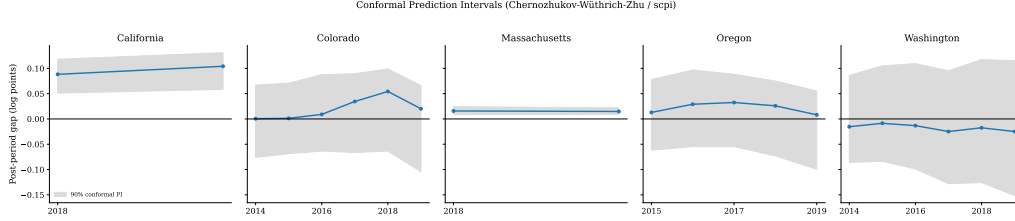


FIGURE 5. PREDICTION INTERVALS FOR THE POST-PERIOD GAPS

Notes: Post-period gaps (actual minus synthetic, log points) with 90% prediction intervals (Cattaneo, Feng and Titiunik, 2021), clean-fit states, Gaussian out-of-sample bounds, 200 simulations.

IV. In-Time Placebos

A constructed counterfactual is credible only if it does not find effects where no treatment occurred. Figure 1 showed that the gap between the treated states and the never-legalized average drifts over the sample; if the estimators absorb that drift imperfectly, they will produce nonzero estimates in windows with no treatment, and the size of those estimates is the noise floor against which any real estimate must be judged. This section measures that floor directly.

I run two exercises. The first backdates the pooled design: all five clean-fit states are assigned a fake 2009 treatment on a panel that ends in 2013, entirely inside every state’s pre-treatment period, and every pooled estimator is re-run with its full inference. The second traces out empirical bias distributions: for each clean-fit state, I estimate the fake effect at every feasible backdated adoption year (requiring at least six pre-treatment years and three fake post-treatment years, with data truncated before real treatment) for the classic, augmented, synthetic difference-in-differences, and generalized synthetic control estimators, 39 fake estimates per method. This measures each estimator’s noise in the panel.

Table 5 and Figures 6–7 report the results. One might expect the latent-factor and time-weighted estimators to absorb the regional drift that affects convex-weight synthetic control; they do not. Every pooled estimator produces a negative fake effect on 2009–2013, ranging from -2.2 to -0.9 percent, with matrix completion closest to rejecting ($t = -1.82$). None rejects at the 5 percent level, and the pooled magnitudes are roughly half those of the single-state classic synthetic control placebos, but the pseudo-effects do not disappear under any estimator. The bias distributions show the same thing from the other direction: fake effects center near -0.9 percent with standard deviations of 2.3 to 3.4 percentage points. The pooled result is not specific to the 2009 window: repeating the exercise with

TABLE 5—IN-TIME PLACEBOS: POOLED FAKE TREATMENT IN 2009

Estimator	Pooled fake $T_0 = 2009$			Single-state bias band	
	Fake ATT	(SE)	t	Mean	SD
Partially-pooled ASCM (primary)	-0.0116	(0.0226)	-0.51	-0.0085 [†]	0.0281
Synthetic difference-in-differences	-0.0137	(0.0166)	-0.83	-0.0111	0.0229
Generalized synthetic control (IFE)	-0.0218	(0.0188)	-1.16	-0.0092	0.0340
Matrix completion	-0.0205	(0.0113)	-1.82	—	—
Callaway–Sant’Anna	-0.0090	(0.0122)	-0.74	—	—
Two-way fixed effects	-0.0205	(0.0137)	-1.49	—	—

Notes: Left panel: all five clean-fit states assigned a fake 2009 treatment, panel 2000–2013 (entirely pre-treatment), never-treated donors, full inference for each estimator. Right panel: mean and standard deviation of 39 single-state backdated placebo estimates per method (every feasible fake adoption year per state). The partially-pooled ASCM is a pooled estimator with no single-unit analogue; its cell (†) instead reports the classic single-treated-unit SCM band, the single-unit special case of the augmented family. Dashes mark estimators for which the single-state backdated exercise was not run (matrix completion, Callaway–Sant’Anna, two-way fixed effects).

a fake 2007 adoption on a panel ending in 2011 again gives uniformly negative estimates, between -2.0 and -1.2 percent. In that window Callaway–Sant’Anna rejects the null at the 5 percent level ($t = -2.22$) on data where no treatment occurred, a false positive that anticipates the over-rejection its standard-error test shows in the power simulation of Section V. Any true effect smaller than two to three percent is inside the range these designs produce when nothing happened. This sets up the question the next section answers: given this noise floor, what effect sizes could the design detect?

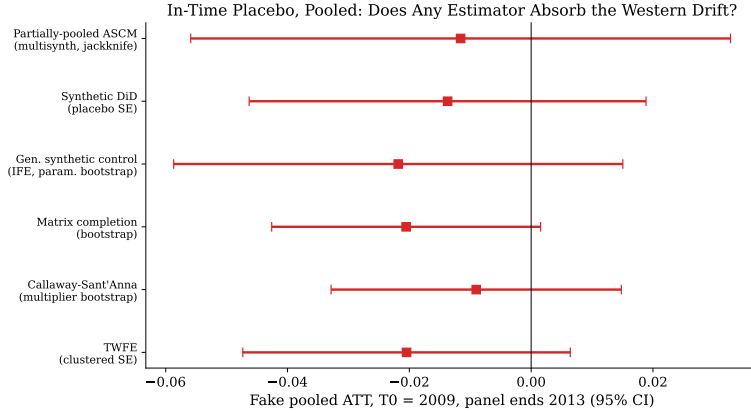


FIGURE 6. POOLED FAKE-2009 PLACEBO ACROSS ESTIMATORS

Notes: Fake pooled ATT with 95% confidence intervals, all clean-fit states assigned $T_0 = 2009$, panel ending 2013.

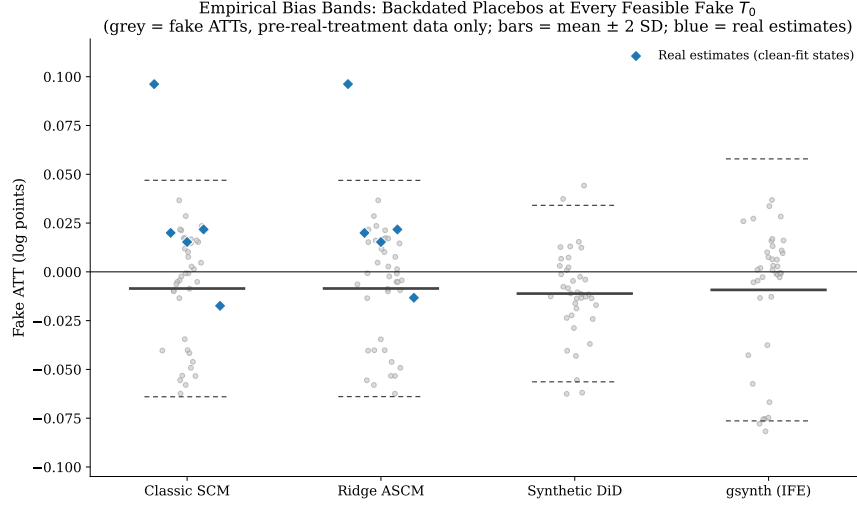


FIGURE 7. EMPIRICAL BIAS BANDS FROM BACKDATED PLACEBOS

Notes: Gray points are fake ATT estimates at every feasible backdated adoption year for every clean-fit state, using only pre-real-treatment data; bars mark the mean and ± 2 SD of each estimator's fake distribution. Blue diamonds are the real estimates for the clean-fit states.

V. Statistical Power and the Minimum Detectable Effect

The placebo evidence gives the design's noise floor. This section asks the logical next question: if substitution of a given size were real, how often would this design detect it?

A. Simulation Design

The simulation preserves the actual design in every dimension. Each draw selects 5 pseudo-treated states at random from the 30 never-treated donors, assigns them the real staggered retail years (2014, 2014, 2015, 2018, 2018) in random order, and injects a multiplicative effect δ into their post-treatment outcomes ($\ln y \leftarrow \ln y + \ln(1 + \delta)$ for $t \geq T_0$). Drawing pseudo-treated states from the never-treated pool guarantees that their outcomes contain no true treatment effect, so any rejection at $\delta = 0$ is a false positive and the rejection rate at $\delta < 0$ is power.⁶ Each estimator is then re-run exactly as in Section III. The grid $\delta \in \{0, -2, -5, -8, -12\}$ percent spans the literature, from the low-single-digit

⁶The simulated power is an upper bound on the actual design's power, so the detectability conclusion is conservative. The pseudo-treated states are drawn from the never-treated donors, which are less drift-prone than the Western states that actually legalized (Figure 1), so the simulation likely understates the noise the real estimator faces. The donor draws also fit their synthetic controls better than the real treated states do, even after the fit screen of Section II.A keeps only the cleanest of those states, so the injected effect sits on top of less pre-existing prediction error. Both features push the reported power above what the real design achieves.

effects that the survey and smaller quasi-experimental evidence suggest to the twelve percent that Baggio, Chong and Kwon (2020) estimate from scanner data. The primary estimator receives the grid with 400 draws per point and the four alternatives with 200, so every estimator gets a power curve and a minimum detectable effect (Bloom, 1995). A final cell repeats the -5 percent case with the effect phasing in linearly over three years, because a constant effect from the opening year is the most detectable shape and the first-stage evidence shows retail markets ramp up gradually.

Two rejection rules are evaluated. The first is each estimator’s inference as conventionally used, which for the primary estimator is the jackknife t -test. The second is a design-calibrated randomization test that rejects when the absolute estimate exceeds the 95th percentile of the same estimator’s $\delta = 0$ distribution. The second rule has exact 5 percent size by construction, so its power column isolates the estimator’s signal-to-noise from any miscalibration of its standard errors.

TABLE 6—DESIGN-BASED POWER OF THE PRIMARY ESTIMATOR

True δ	Draws	Reject (jackknife)	Reject (RI-calibrated)	Mean estimate
0%	400	0.0%	5.0%	0.0036
-2%	400	0.0%	10.5%	-0.0166
-5%	400	0.0%	63.7%	-0.0477
-8%	400	0.0%	98.0%	-0.0797
-12%	400	0.0%	100.0%	-0.1242
-5% (3-yr phase-in)	400	0.0%	29.8%	-0.0319

Notes: Partially-pooled augmented synthetic control on pseudo-treated never-treated states with the real staggered dates. Rejection columns report the share of draws rejecting at the 5 percent level: the jackknife rule rejects when $|ATT/SE| > 1.96$, and the RI-calibrated rule rejects when $|ATT|$ exceeds the 95th percentile of the estimator’s own $\delta = 0$ draws (size 5% by construction). The mean-estimate column shows that the estimator recovers the injected effect nearly without bias. In the final row the -5 percent effect phases in linearly over three years ($\delta/3$, $2\delta/3$, then full) instead of arriving at once.

B. Results

Table 6 contains two findings. The first concerns the inference. The mean estimate tracks the injected effect almost exactly (-0.048 against a true -0.050 ; -0.080 against -0.080), so the estimator is approximately unbiased in this design. The jackknife test, however, rejects in 0.0 percent of draws at $\delta = 0$ and at every other value of δ : zero rejections in 400 draws even when the true effect is -12 percent, three to four times the low-single-digit effects the survey evidence implies and at the high end of anything the literature reports. The failure sits in the standard error rather than the point estimate: at $\delta = 0$ the mean jackknife standard error is 1.7 times the true sampling dispersion of the estimator, and the inflation grows with the injected effect, reaching 6.5 times at $\delta = -12$ percent, so the t -statistic stays below 1.96 at every effect size. With five treated units, a

jackknife-based null from this estimator carries close to no evidential weight, here or in other applications of the method at this scale.

The second finding concerns the data. Under the correctly sized randomization rule, the primary estimator’s power is 10 percent against a 2 percent effect, 64 percent against 5 percent, 98 percent against 8 percent, and 100 percent against 12 percent (Figure 8). Interpolating, its minimum detectable effect at 80 percent power is 6.4 percent of per-capita ethanol. At the clean-fit states’ 2013 baseline of 2.84 gallons of pure ethanol per adult, that is roughly 3.2 standard drinks per adult per month. Table 7 reports the same grid for the alternatives. The standard-error tests of synthetic difference-in-differences, Callaway–Sant’Anna, and matrix completion over-reject at $\delta = 0$ (empirical sizes of 8, 10, and 12 percent against a nominal 5), so their apparent power partly reflects miscalibration rather than signal. Under the calibrated thresholds the most sensitive estimator in the set is Callaway–Sant’Anna, with a minimum detectable effect of 4.5 percent. That figure is the design’s limit across the estimators considered, and it is the relevant benchmark for what these methods can show on this data. Detection fails asymmetrically across the range of effects contained in the literature.⁷ An effect of 12 percent, the high end and the magnitude Baggio, Chong and Kwon (2020) report, would be caught by every calibrated test. An effect of 2 percent is far below every threshold: the most sensitive estimator rejects it in 26 percent of draws and the primary estimator in 10 percent. An effect of 4 percent reaches the detectability frontier of the best estimator alone, caught about 70 percent of the time, short of the 80 percent convention, while the primary estimator detects it in 46 percent. The low single digits, where the survey evidence and the smaller quasi-experimental estimates point, are therefore at or below what this design can see. The phase-in row of Table 6 adds a caution in the same direction: a -5 percent effect that ramps in over three years is detected in 30 percent of draws, against 64 percent for the constant version, so a realistic adoption path lowers power further.

VI. Robustness

Table 8 re-estimates the primary specification varying one ingredient at a time. The pooled null is stable across all variations.

Spillovers and contamination. Donors adjacent to any treated state are exposed to cross-border cannabis purchasing (Hansen, Miller and Weber, 2020), and high-intensity medical-cannabis states are partially treated in substance if not in law.

⁷The literature does not provide a single aggregate elasticity to compare against. The quasi-experimental estimates span null effects (Veligati et al., 2020; Armstrong et al., 2025), mixed beverage- and state-specific results (Calvert and Erickson, 2021), and the 12 percent retail decline of Baggio, Chong and Kwon (2020), with Anderson, Hansen and Rees (2013) finding substitution concentrated in beer and among young adults. Survey studies report that a large share of cannabis users reduce their drinking (Subbaraman, 2016), but user-level substitution rates need not aggregate to large population effects once cannabis-use prevalence and intensity are accounted for. I therefore treat the low single digits as the relevant small-effect case rather than as a point estimate from any one study.

TABLE 7—POWER AND MINIMUM DETECTABLE EFFECTS BY ESTIMATOR

Estimator	Size, SE test	Power, RI-calibrated test				MDE
	$\delta = 0$	−2%	−5%	−8%	−12%	
Partially-pooled ASCM (primary)	0.0%	10.5%	63.7%	98.0%	100.0%	6.4%
Synthetic difference-in-differences	8.0%	17.5%	85.5%	100.0%	100.0%	4.8%
Generalized synthetic control (IFE)	3.5%	14.5%	71.0%	98.5%	99.5%	6.0%
Matrix completion	12.0%	12.5%	65.5%	96.0%	100.0%	6.4%
Callaway–Sant’Anna	10.0%	26.0%	92.0%	100.0%	100.0%	4.5%

Notes: Entries are the share of draws rejecting at the 5 percent level; 400 draws per cell for the primary estimator, 200 for the alternatives. The size column uses each estimator’s native standard errors (with reduced replication counts inside the simulation: 25 placebo replications per cohort for synthetic difference-in-differences, 100 bootstrap draws for the generalized synthetic control and matrix completion, 199 multiplier iterations for Callaway–Sant’Anna). The power columns use each estimator’s own $\delta = 0$ 95th percentile as the rejection threshold, which has 5% size by construction. The MDE column interpolates the calibrated power curve to 80 percent power.

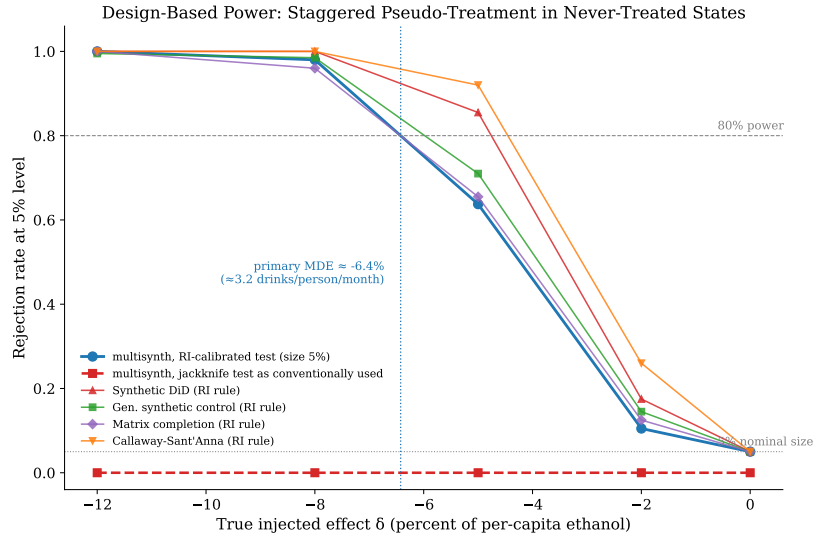


FIGURE 8. POWER CURVES AND THE MINIMUM DETECTABLE EFFECT

Notes: Rejection rates at the 5% level against the injected effect δ . The heavy solid curve is the primary estimator under the design-calibrated randomization test; the dashed curve is the same estimator under its jackknife test, which never leaves zero. The thin curves are the four alternative estimators under their own calibrated thresholds. The dotted vertical line marks the primary estimator’s minimum detectable effect at 80 percent power.

TABLE 8—ROBUSTNESS OF THE POOLED ESTIMATE

Specification	ATT	(SE)	Percent
Primary specification	0.0276	(0.0419)	+2.8
Drop donors adjacent to treated states	0.0289	(0.0397)	+2.9
Drop Oklahoma (high-intensity medical)	0.0276	(0.0411)	+2.8
Drop New Hampshire (cross-border sales)	0.0276	(0.0431)	+2.8
Timing-adjusted donor pool	0.0254	(0.0464)	+2.6
Extension window, 2000–2023	0.0416	(0.0627)	+4.2
All seven treated states	0.0225	(0.0318)	+2.3
Beer ethanol only	0.0385	(0.0521)	+3.9
Wine ethanol only	-0.0295	(0.0691)	-2.9
Spirits ethanol only	0.0340	(0.0490)	+3.5
Washington, spirits-excluded outcome	-0.0289	[-0.100, +0.038]	-2.9

Notes: All rows except the last are the partially-pooled augmented synthetic control with jackknife standard errors, varying one ingredient at a time from the primary specification. The Washington spirits-excluded row is a single-state ridge-augmented estimate on beer-plus-wine ethanol with a jackknife+ 95% interval in place of a standard error.

Dropping the adjacent donors, and separately dropping Oklahoma, whose post-2018 medical program produced the country’s highest dispensary density, moves the pooled estimate by less than a tenth of a standard error. So does dropping New Hampshire, the donor whose state-run, untaxed liquor sales attract the heaviest cross-border purchasing and which carries large weights in several synthetic controls (Appendix Table A1). The direction of any remaining contamination bias is known: under true substitution, contaminated donors drink less, which biases the treated-state gaps upward, against finding substitution, and is the likely source of the uniformly positive point estimates. This is another reason the positive California gap should not be read causally.

Donor pool and window. The timing-adjusted donor pool (admitting states that legalized after 2019) and the full 2000–2023 extension window (which adds the COVID years, +4.2 percent) leave the conclusion unchanged, as does re-admitting Alaska and Nevada in defiance of the fit screen.

Beverage decomposition. Pooled estimates by beverage, with jackknife standard errors, give positive nulls for beer and spirits and a negative estimate for wine (−2.9 percent). Wine is the only beverage whose sign matches substitution, and its magnitude sits inside the range the power analysis shows this design cannot distinguish from zero. That is the paper’s point, just on a smaller scale: a substitution-signed estimate of plausible size is exactly what an underpowered design produces by chance.

Washington and privatization. On the spirits-excluded outcome, Washington’s estimate is −2.9 percent with an interval of [−0.100, +0.038] log points. This sign matches substitution, but the interval covers zero comfortably.

Covariates. The lag-subset specification (odd-year lags plus four pre-treatment covariates, each predictor standardized across donors) yields state-level estimates more dispersed than the outcome-only match but with the same pooled message

(Appendix Table B1). The added dispersion is consistent with Kaul et al. (2022): covariate matching changes the appearance of the specification without improving it.

VII. Discussion

Three results structure what this paper can and cannot say. The pooled estimates show that the aggregate null holds across identification approaches, so estimator choice does not explain the literature’s aggregate non-findings. The placebo analysis shows that every one of those estimators produces one-to-two percent pseudo-effects on no-treatment data in this setting, so estimates of that magnitude, at the small end of the low single digits where plausible substitution would lie, are inside the design’s noise floor. The power analysis makes the same point: even with correctly calibrated inference, no estimator in the set can detect effects smaller than 4.5 percent at conventional power, and the inference actually attached to the primary estimator in applied work detects nothing at all.

The implication for the literature follows. Aggregate-sales nulls are evidence about detectability, not evidence that cannabis–alcohol substitution is absent. The granular studies that find substitution (Baggio, Chong and Kwon, 2020; Anderson, Hansen and Rees, 2013), the mixed household-purchase results of Calvert and Erickson (2021), and the aggregate studies that find nothing (Veligati et al., 2020; Armstrong et al., 2025) are consistent with each other: a 12 percent scanner-data effect concentrated in specific products and places is compatible with an aggregate effect in the low single digits that no state-panel design of this size can see. The policy implication is that claims that legalization measurably reduces population drinking, and claims that it measurably does not, both outrun what this class of evidence can support.

The methodological point is the more portable of the two. Staggered synthetic-control designs with a handful of treated units are now common, and their null results are routinely reported with jackknife or placebo-based uncertainty. The simulation here shows that at least one standard configuration of that machinery (five treated units, thirty donors, twenty annual periods) pairs a nearly unbiased estimator with a test of essentially zero size and zero power, so a null from it was never capable of saying otherwise. A design-based power statement of the kind in Section V is cheap to produce and should accompany null results when this research design is used.

The limitations are the familiar ones. The outcome measures sales, not consumption, and attributes cross-border purchases to the selling state. The treatment is binary retail opening, while effective dose varies by an order of magnitude across states; the first-stage section documents this but the pooled estimators do not exploit it, and a dose-response design is a natural next step. The donor pool is plausibly contaminated by national cannabis diffusion, with a known anti-substitution bias direction but unknown magnitude. The power simulation injects multiplicative effects that are constant or follow a simple three-year phase-in;

richer dynamics would change the numbers, though the phase-in result suggests that added realism lowers power rather than raising it, which leaves the conclusion that effects in the low single digits are out of reach intact.

VIII. Conclusion

This paper estimates the effect of recreational cannabis retail openings on state-level alcohol sales and asks what such a design can actually detect. The findings are consistent across three exercises. Six estimators return a band of small, positive-signed nulls. Backdated placebos show that every estimator produces spurious effects of the same size as the substitution it is meant to find. And a power simulation shows that the primary estimator’s standard inference rejects nothing at any effect size, while even correctly calibrated inference cannot detect effects below 4.5 percent of per-capita ethanol.

The nulls are therefore not evidence against substitution. What two decades of state-level sales data can establish is narrow but real: they rule out large aggregate effects, including the twelve-percent-scale decline found in scanner data, but they cannot resolve the low-single-digit effects that plausible substitution would produce. On that question the data are silent, not negative.

The lesson extends past cannabis. A small-sample staggered synthetic control reported with jackknife inference is close to guaranteed to find nothing, whatever the truth, and only a power statement reveals whether its null carries any information. Detecting effects of plausible size will require what the studies that find substitution already use: granular outcomes, credible dose variation, and designs whose power is demonstrated rather than assumed.

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DONOR WEIGHTS AND COVARIATE BALANCE

TABLE A1—SYNTHETIC CONTROL DONOR WEIGHTS

Treated state	Donors (weight)
Alaska	Texas (0.661), New Hampshire (0.328), Kansas (0.011)
California	Florida (0.271), Minnesota (0.195), Georgia (0.133), Utah (0.116), South Carolina (0.109), Louisiana (0.102), North Dakota (0.032), Tennessee (0.029)
Colorado	Georgia (0.346), New Hampshire (0.232), Minnesota (0.173), Wisconsin (0.102), Louisiana (0.072), Florida (0.063), North Carolina (0.012)
Massachusetts	South Carolina (0.192), Texas (0.150), New Hampshire (0.135), Georgia (0.131), Florida (0.126), Hawaii (0.090), Tennessee (0.081), Louisiana (0.054)
Nevada	New Hampshire (0.668), Georgia (0.332)
Oregon	Hawaii (0.278), Louisiana (0.234), North Dakota (0.180), Kentucky (0.100), Iowa (0.075), Florida (0.052), Delaware (0.048), Wyoming (0.033)
Washington	South Carolina (0.390), Hawaii (0.325), Kansas (0.266), Minnesota (0.019)

Notes: Classic synthetic control weights (outcome-only match, never-treated donors, 2000–2019), donors with weight above 0.001, largest eight per state.

TABLE A2—COVARIATE BALANCE, PRE-TREATMENT

State (T_0)	Covariate	Treated	Synthetic	Donor average
Colorado (2014)	GDP per capita (\$)	56,117	52,998	51,824
	Unemployment (%)	7.8	7.2	6.8
	Share aged 20–34	0.214	0.198	0.202
	Beer tax (\$/gal)	0.08	0.27	0.29
Washington (2014)	GDP per capita (\$)	61,728	49,513	51,824
	Unemployment (%)	8.2	7.0	6.8
	Share aged 20–34	0.209	0.199	0.202
	Beer tax (\$/gal)	0.76	0.65	0.29
Oregon (2015)	GDP per capita (\$)	48,579	58,876	52,601
	Unemployment (%)	7.7	5.3	6.1
	Share aged 20–34	0.205	0.209	0.203
	Beer tax (\$/gal)	0.08	0.42	0.30
Alaska (2016)	GDP per capita (\$)	73,024	58,017	53,391
	Unemployment (%)	6.6	5.0	5.4
	Share aged 20–34	0.224	0.207	0.203
	Beer tax (\$/gal)	1.07	0.23	0.31
Nevada (2017)	GDP per capita (\$)	52,547	56,994	54,017
	Unemployment (%)	7.0	4.4	4.8
	Share aged 20–34	0.207	0.193	0.203
	Beer tax (\$/gal)	0.16	0.31	0.32
California (2018)	GDP per capita (\$)	67,376	53,414	54,436
	Unemployment (%)	5.5	4.6	4.3
	Share aged 20–34	0.221	0.205	0.202
	Beer tax (\$/gal)	0.20	0.41	0.33
Massachusetts (2018)	GDP per capita (\$)	76,441	53,523	54,436
	Unemployment (%)	4.2	4.5	4.3
	Share aged 20–34	0.213	0.201	0.202
	Beer tax (\$/gal)	0.11	0.53	0.33

Notes: Covariates measured as each state’s average over the three years before its own treatment year. Synthetic values use the classic outcome-only weights of Table A1; balance is therefore diagnostic (covariates are not matched on in the primary specification, per Section II).

ADDITIONAL ROBUSTNESS TABLES

TABLE B1—LAG-SUBSET SPECIFICATION WITH PRE-TREATMENT COVARIATES

State	T_0	Outcome-only ATT	Lag-subset + covariates ATT
Colorado	2014	0.0200	0.0212
Washington	2014	-0.0174	-0.0697
Oregon	2015	0.0217	0.0629
California	2018	0.0962	0.1057
Massachusetts	2018	0.0153	-0.0335

Notes: Classic synthetic control per clean-fit state. The outcome-only column matches on all pre-treatment lags; the lag-subset column matches on odd-year lags plus real GDP per capita, unemployment, the share aged 20–34, and the beer excise tax, each measured as the average over the three years before the state’s treatment and standardized across donors (Kaul et al., 2022). ATT in log points.

TABLE B2—FIRST-STAGE DOSE BY STATE

State	Retail year	Sales per adult, first two years (\$)
Nevada	2017	236
Colorado	2014	213
California	2018	83
Massachusetts	2018	43
Washington	2014	34

Notes: Mean annual legal cannabis sales per adult aged 21 and older over each state’s first two retail years, nominal dollars, from state tax and regulatory sources. Oregon is omitted: its agency publishes only rounded totals beginning in 2020, after its retail opening. Washington and Nevada report fiscal years. California’s figure is taxable sales.